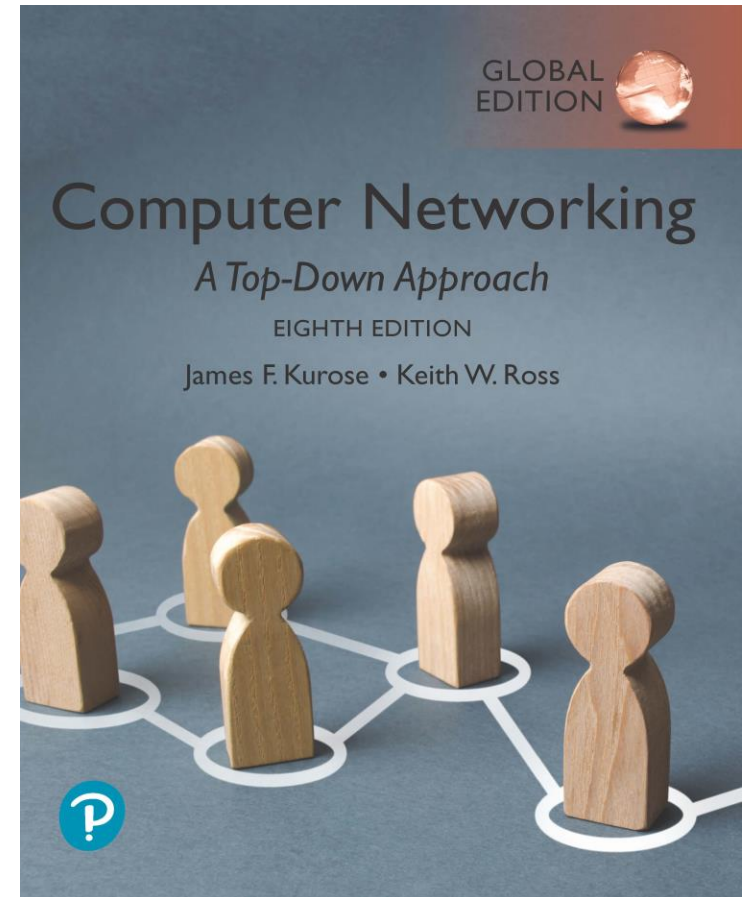


Chapter 5

Network Layer: Control Plane



Computer Networking: A Top-Down Approach

8th Edition, Global Edition

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Network layer control plane: our goals

- understand principles behind network control plane:
 - traditional routing algorithms
- instantiation, implementation in the Internet:
 - Internet Control Message Protocol: ICMP

Network layer: “control plane” roadmap

- introduction
- routing protocols
 - Internet Control Message Protocol



Network-layer functions

- **forwarding**: move packets from router's input to appropriate router output
- **routing**: determine route taken by packets from source to destination

data plane

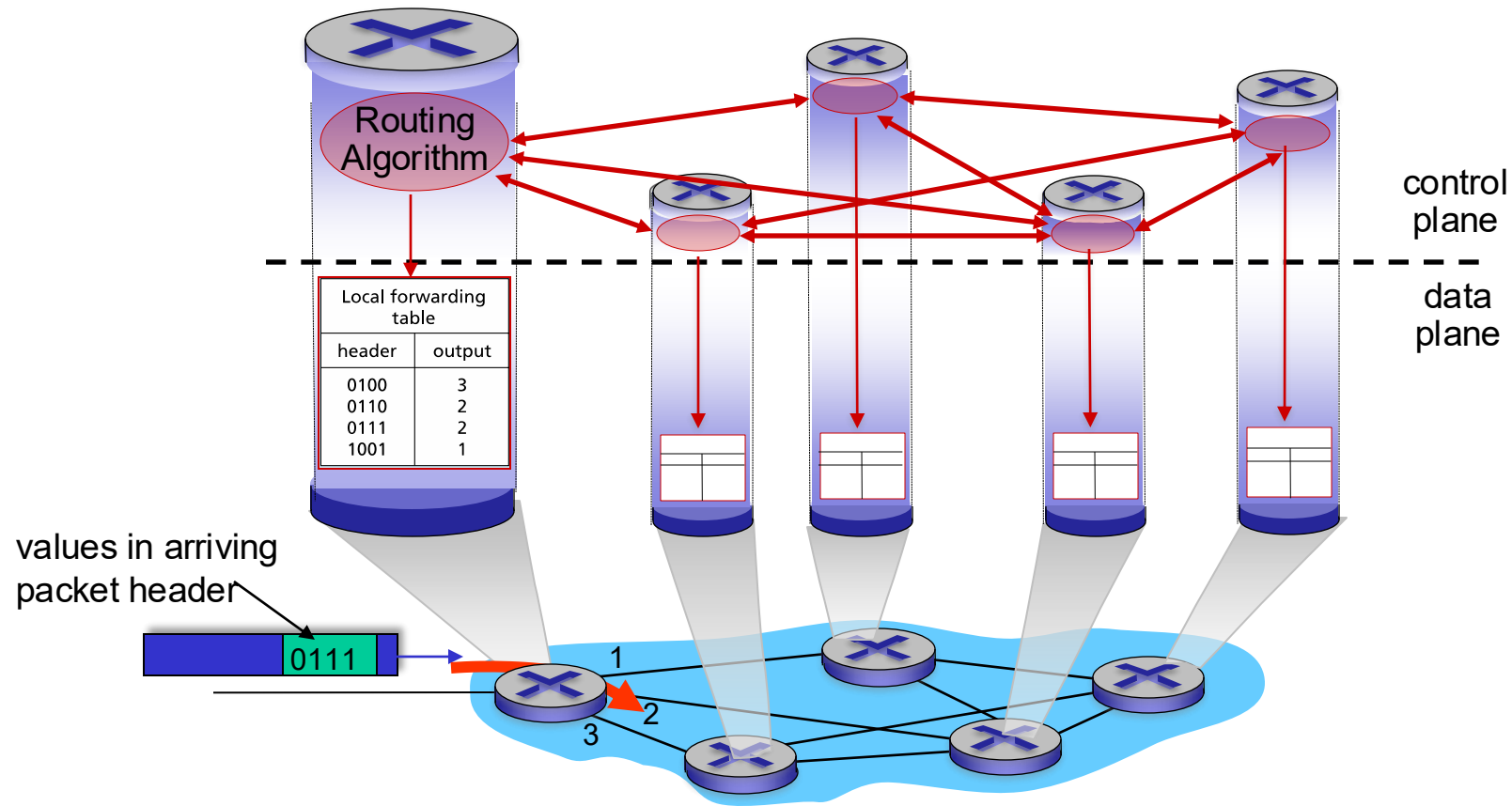
control plane

Two approaches to structuring network control plane:

- per-router control (traditional)
- logically centralized control (software defined networking)

Per-router control plane

Individual routing algorithm components *in each and every router* interact in the control plane



Network layer: “control plane” roadmap

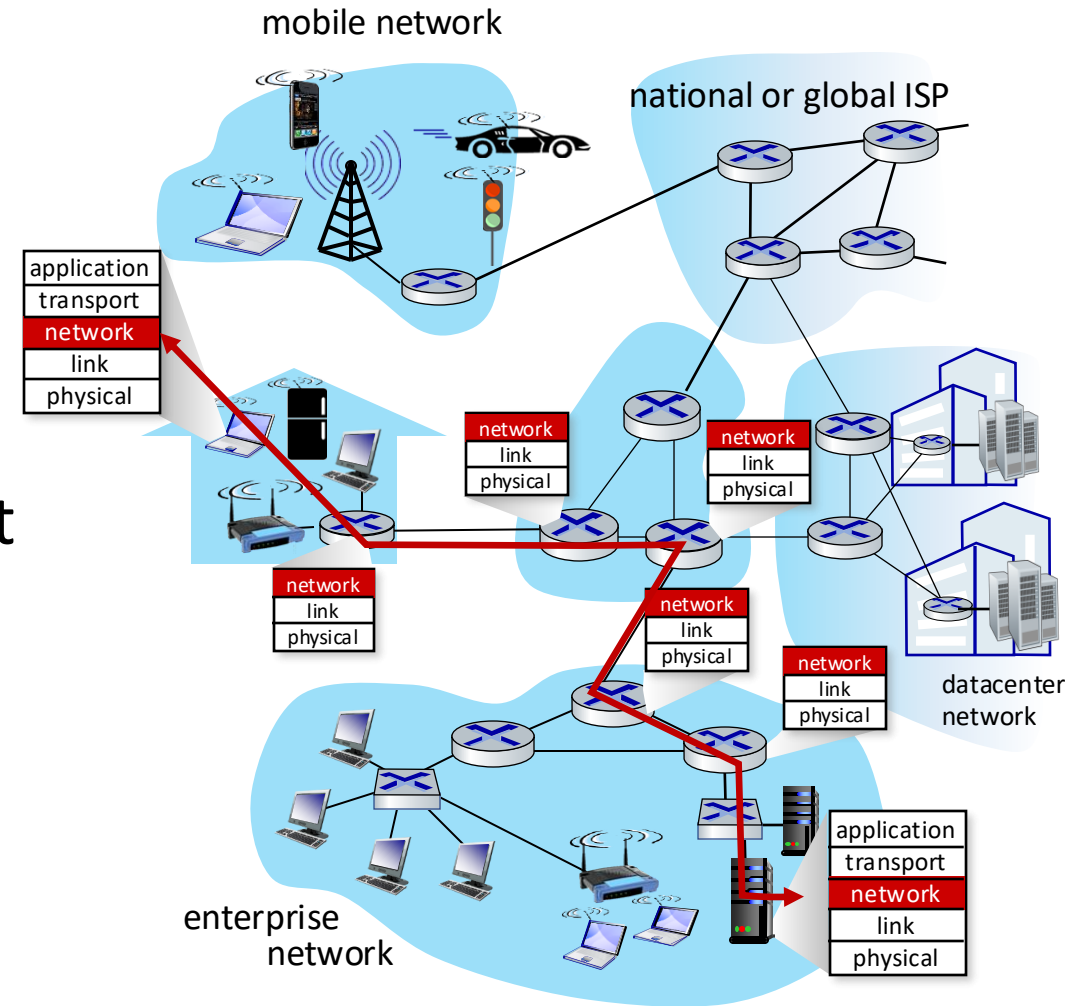
- introduction
- **routing protocols**
- Internet Control Message Protocol



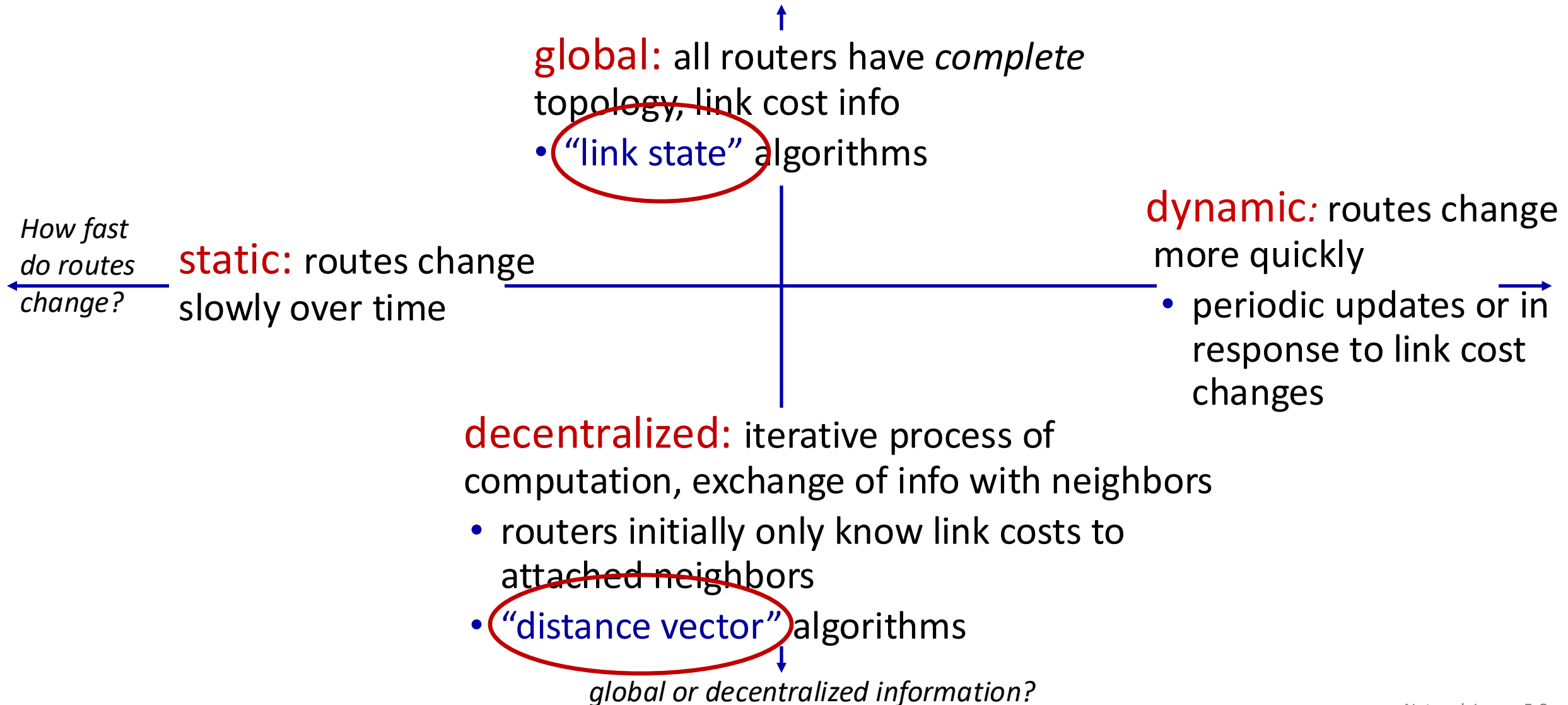
Routing protocols

Routing protocol goal: determine “good” paths (equivalently, routes), from sending hosts to receiving host, through network of routers

- **path:** sequence of routers packets traverse from given initial source host to final destination host
- **“good”:** least “cost”, “fastest”, “least congested”
- routing: a “top-10” networking challenge!



Routing algorithm classification





Her er en forklaring på "distance vector routing protocol" på norsk, kombinert med det viktigste fra gode engelske forklaringer:

Hva er en "Distance Vector Routing Protocol"?

En "distance-vector routing protocol" er en metode som rutere i datanettverk bruker for å finne den beste ruten for å sende datapakker. Den fokuserer primært på to ting:

- **Distanse:** Antall rutere (hopp) en pakke må passere for å nå målet. Kortere distanse anses som bedre.
- **Vektor:** Retningen pakken må sendes i for å nå målet via den korteste ruten.

Hvordan fungerer det?

1. **Informasjonsdeling:** Hver ruter deler sin egen avstandstabell (en liste over kjente destinasjoner og avstander til dem) med sine direkte nabo-rutere.
2. **Oppdatering:** Ruterne bruker naboens informasjon til å oppdatere sin avstandstabell. Hvis en nabo presenterer en kortere rute til en destinasjon, oppdaterer ruterne for å benytte seg av denne.
3. **Konvergens:** Denne prosessen med å dele og oppdatere informasjon fortsetter til alle rutere i nettverket har en konsekvent forståelse av de optimale rutene.

Hva er en "link-state routing protocol"?

En link-state routing protocol er en annen metode som rutere bruker for å dynamisk bestemme de beste rutene i et datanettverk. I motsetning til distance-vector protokoller, fokuserer link-state protokoller på tilstanden til koblingene i nettverket.

Kjennetegn på Link-State protokoller:

- **Kartlegging av nettverket:** Hver ruter bygger et komplett kart over nettverkstopologien, og inkluderer informasjon om hver eneste ruter og koblingene mellom dem.
- **Link-state pakker (LSPs):** Rutere sender ut små oppdateringspakker kalt "Link-State Advertisements" (LSAs) eller "Link-State Pakker" (LSPs) som beskriver direkte tilknyttede koblinger og tilhørende kostnader (metrikk).
- **Flooding:** Disse LSA-ene blir 'flooded' gjennom nettverket slik at alle rutere mottar den samme informasjonen.
- **Kalkulering av beste rute:** Hver ruter bruker sin kunnskap om hele nettverket til å uavhengig kalkulere de korteste stiene til alle mulige destinasjoner.

Fordeler med Link-State protokoller:

- **Rask konvergens:** Endringer i nettverket spres raskt, og rutere oppdaterer routingstabellene sine kjapt.
- **Tilpasningsevne:** Passer bra for store og komplekse nettverk.
- **Bedre skalerbarhet:** Håndterer endringer i nettverkstopologien mer effektivt sammenlignet med distance-vector protokoller.

Network layer: “control plane” roadmap

- introduction
- routing protocols
- **Internet Control Message Protocol**

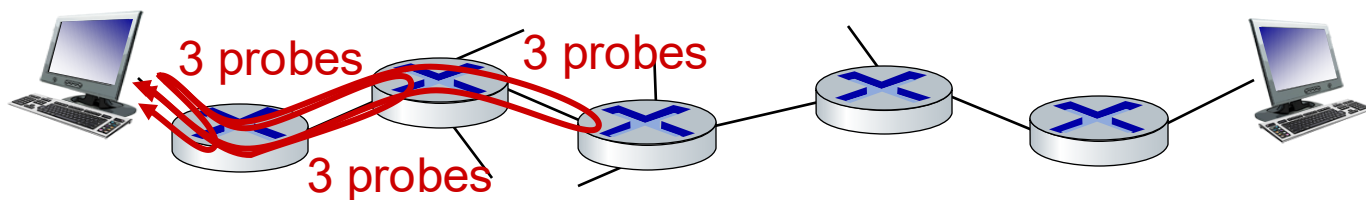


ICMP: internet control message protocol

- used by hosts and routers to communicate network-level information
 - error reporting: unreachable host, network, port, protocol
 - echo request/reply (used by ping)
- network-layer “above” IP:
 - ICMP messages carried in IP datagrams
- *ICMP message*: type, code plus first 8 bytes of IP datagram causing error

<u>Type</u>	<u>Code</u>	<u>description</u>
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

Traceroute and ICMP



Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

- source sends sets of UDP segments to destination
 - 1st set has TTL =1, 2nd set has TTL=2, etc.
- datagram in n th set arrives to n th router:
 - router discards datagram and sends source ICMP message (type 11, code 0) **TTL expired**
 - ICMP message possibly includes name of router & IP address
- when ICMP message arrives at source: record RTTs

stopping criteria:

- UDP segment eventually arrives at destination host
- destination returns ICMP “port unreachable” message (type 3, code 3)
- source stops

What is network management?

- autonomous systems (aka “network”): 1000s of interacting hardware/software components
- other complex systems requiring monitoring, configuration, control:
 - jet airplane, nuclear power plant, others?



"**Network management** includes the deployment, integration and coordination of the hardware, software, and human elements to monitor, test, poll, configure, analyze, evaluate, and control the network and element resources to meet the real-time, operational performance, and Quality of Service requirements at a reasonable cost."